PYTHON PROJECT

MANAGEMENT AND INFO



PROJECT MANAGEMENT

PYTHON PROJECT

GROUP PROJECT: THE SCHEDULE (SECTIONS 1,4)

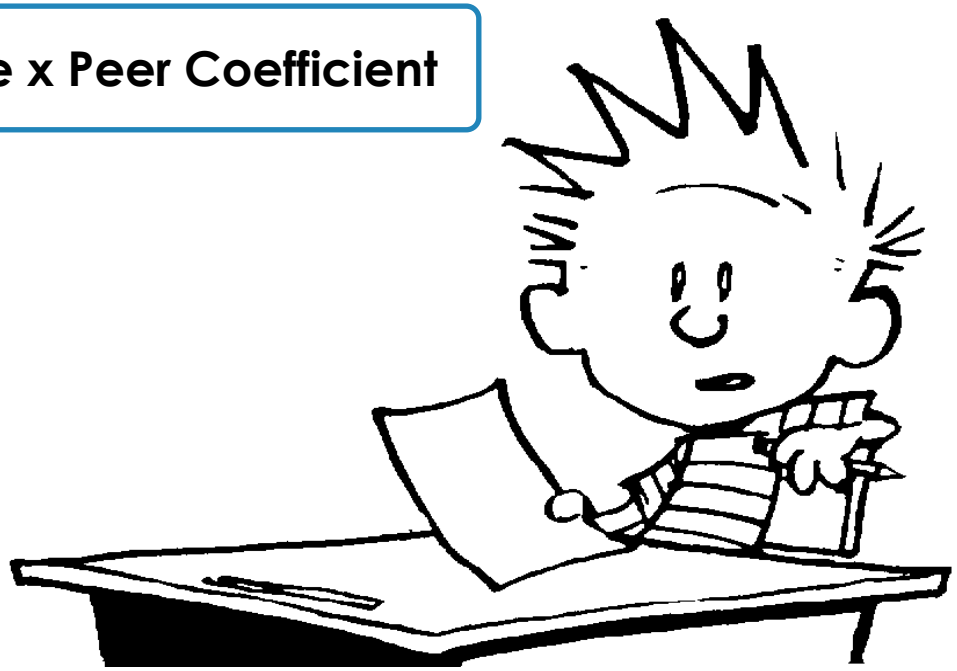
Week	Date	Pdf #	Topics	Assignments/ Due Dates
12	11-09	P	Project Overview	**Project assigned**
	11-11	P	Project Discussion Time	
13	11-16	P	**Hackathon 3 (quiz)** Project Discussion Time	**HW4 due**
	11-18	P	Project Discussion Time	
14	11-23		No class	
	11-25		No class	
15	11-30	P	Project Discussion Time	
	12-02	P	**Hackathon 4 (quiz)** Project Final Discussion Time	
16	**12-07**		Project Presentations	**Project Due**
	12-09	R	Review + Pizza	
17	**12-16** **12-16**		**FINAL EXAM (quiz) – Section 1 (7:30 - 9:30 AM)** **FINAL EXAM (quiz) – Section 4 (12:00 - 2:00 PM)**	

PYTHON PROJECT

GROUP PROJECT: THE GRADE

- **The team** will collectively develop the **project**
- Your project grades will be given by the **group grade** (team based) multiplied by a **peer coefficient** (individual):

Project Grade = Group Grade x Peer Coefficient

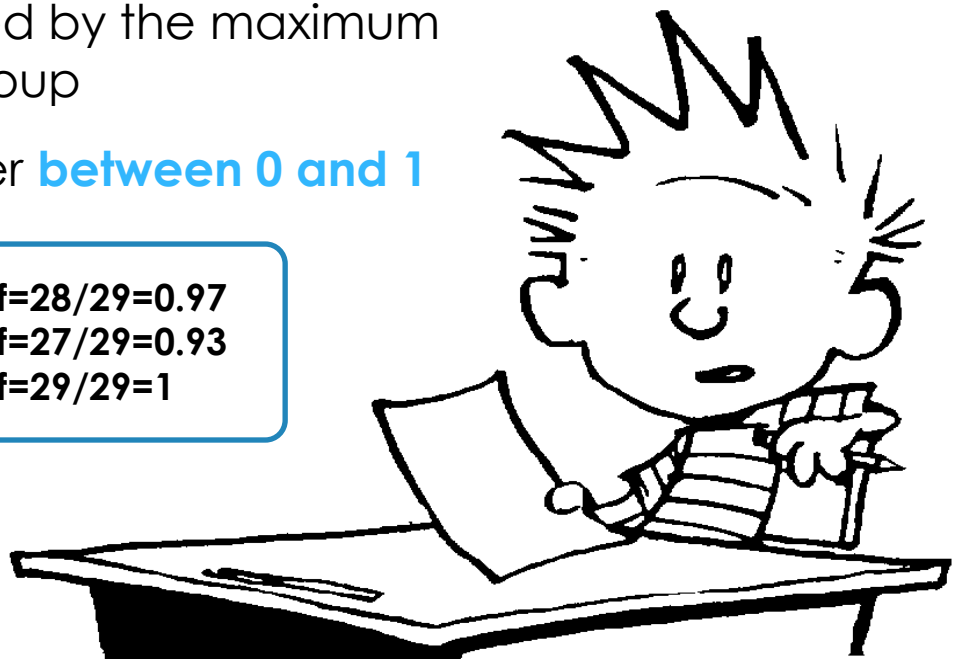


PYTHON PROJECT

GROUP PROJECT: THE GRADE

- You will be required to **evaluate your teammates** and **yourself**
- The **peer coefficient** will be given by your peer grade (sum of the grades that your team members and yourself assigned to you) divided by the maximum peer grade received in your group
- The peer coefficient is a number **between 0 and 1**

Black: 8+10+10=28	**PeerCoeff=28/29=0.97**
White: 8+9+10=27	**PeerCoeff=27/29=0.93**
Brown: 10+9+10=29	**PeerCoeff=29/29=1**



PYTHON PROJECT

WHY WORKING IN TEAMS?

There are many advantages to working in a team:

- learning to **plan** and **organize** individual and group work
- learning to **negotiate** and **compromise**
- practicing **decision-making skills**
- gaining **knowledge** from another person
- **exchanging experiences** with new people
- having **fun** (!)



PYTHON PROJECT

WHY WORKING IN TEAMS?

- Working in groups can be **challenging** when you have **different personalities** on the team, but the collaborative work normally **accomplishes results** that individuals working on their own cannot reach
- Overall, team work provides a **practical real-life experience**: few people will obtain jobs that enable them to work in complete isolation so, among technical skills, it is necessary to learn **how to work** (**efficiently**) **with other people**



PYTHON PROJECT

TEAM NORMS

- **Meet with your team** and exchange **contact information** and **schedules**
- Decide **when and where** you will meet **outside of class**
- Lay out a **schedule** that includes important **deadlines**
- Agree on **rules for the team** (**team norms**) that everyone must follow (e.g.: show up prepared and on time, text the team if you are unable to make a meeting, be professional with one another)



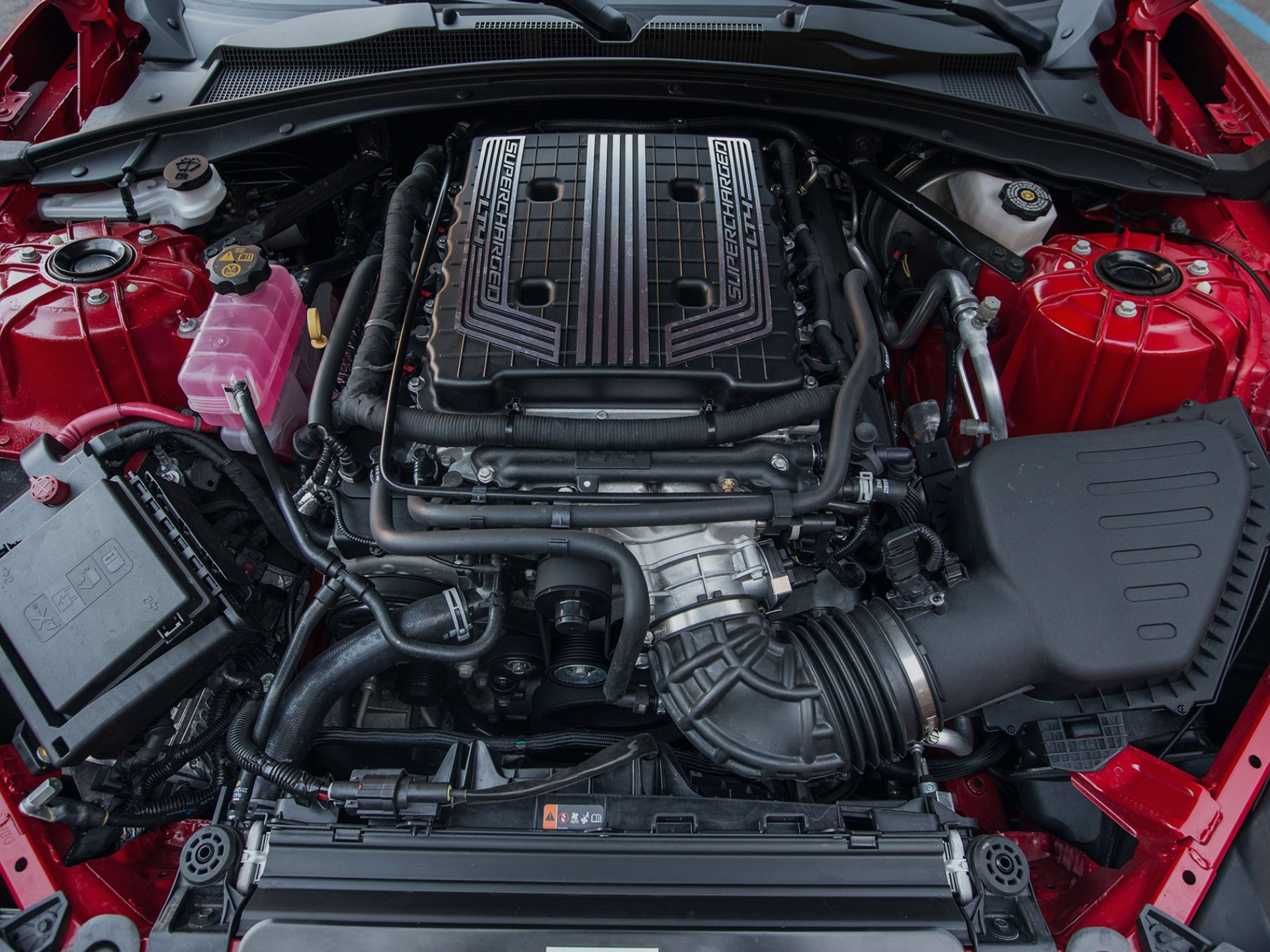
PYTHON PROJECT

TEAM NORMS

- **Team norms** are a set of **guidelines** that teams use to make working together as a team as **effective** and **productive** as possible
- The norms are useful in reminding them that an **agreed upon set of guidelines** exist and that those guidelines **need to be followed**
- If **unprofessional behavior** continues beyond a reminder, the **project manager** (**professor**) will need to be **consulted** and asked to step in

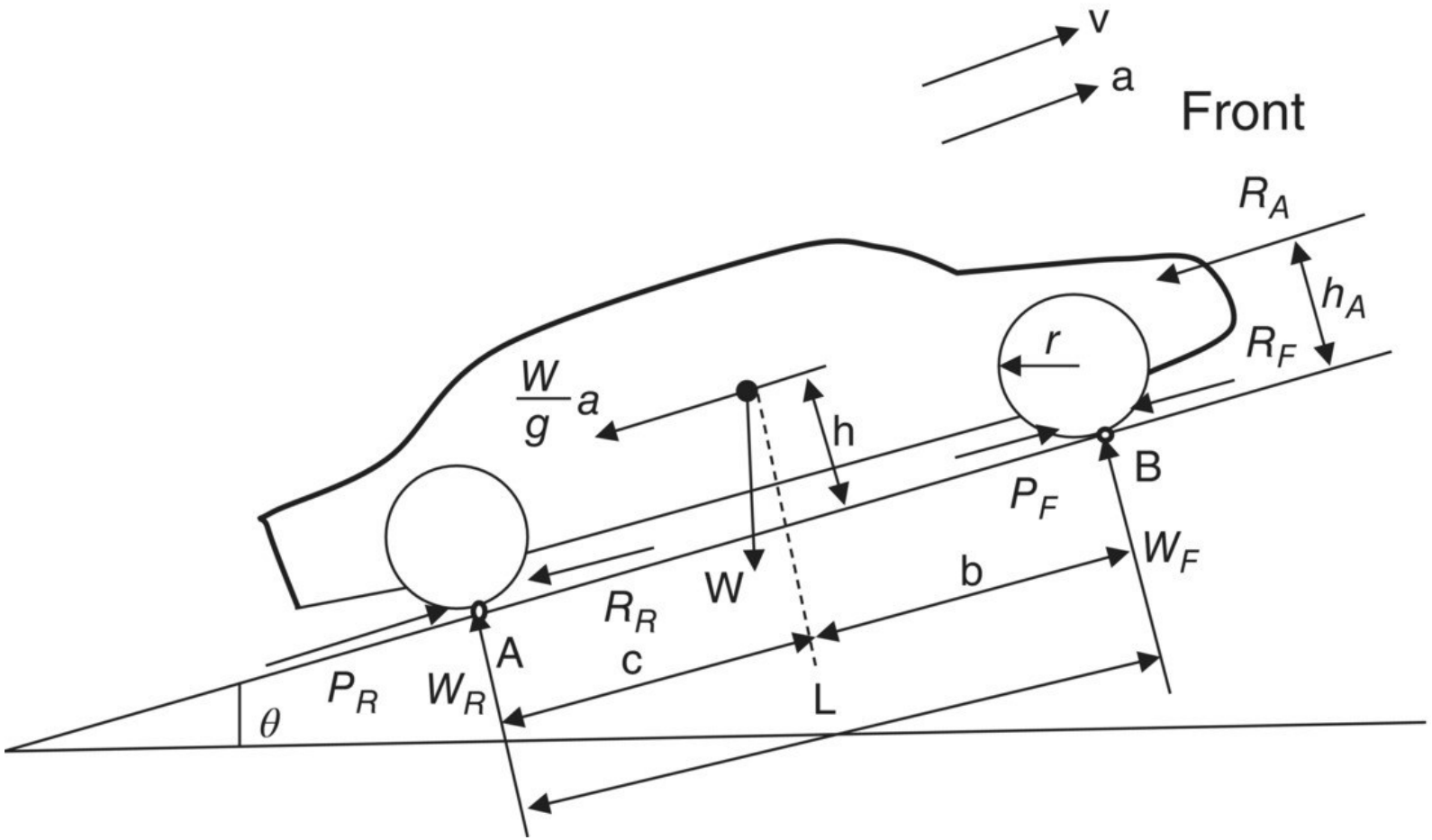


PROJECT TOPIC


SUPERCHARGED
LTH
SUPERCHARGED
LTH

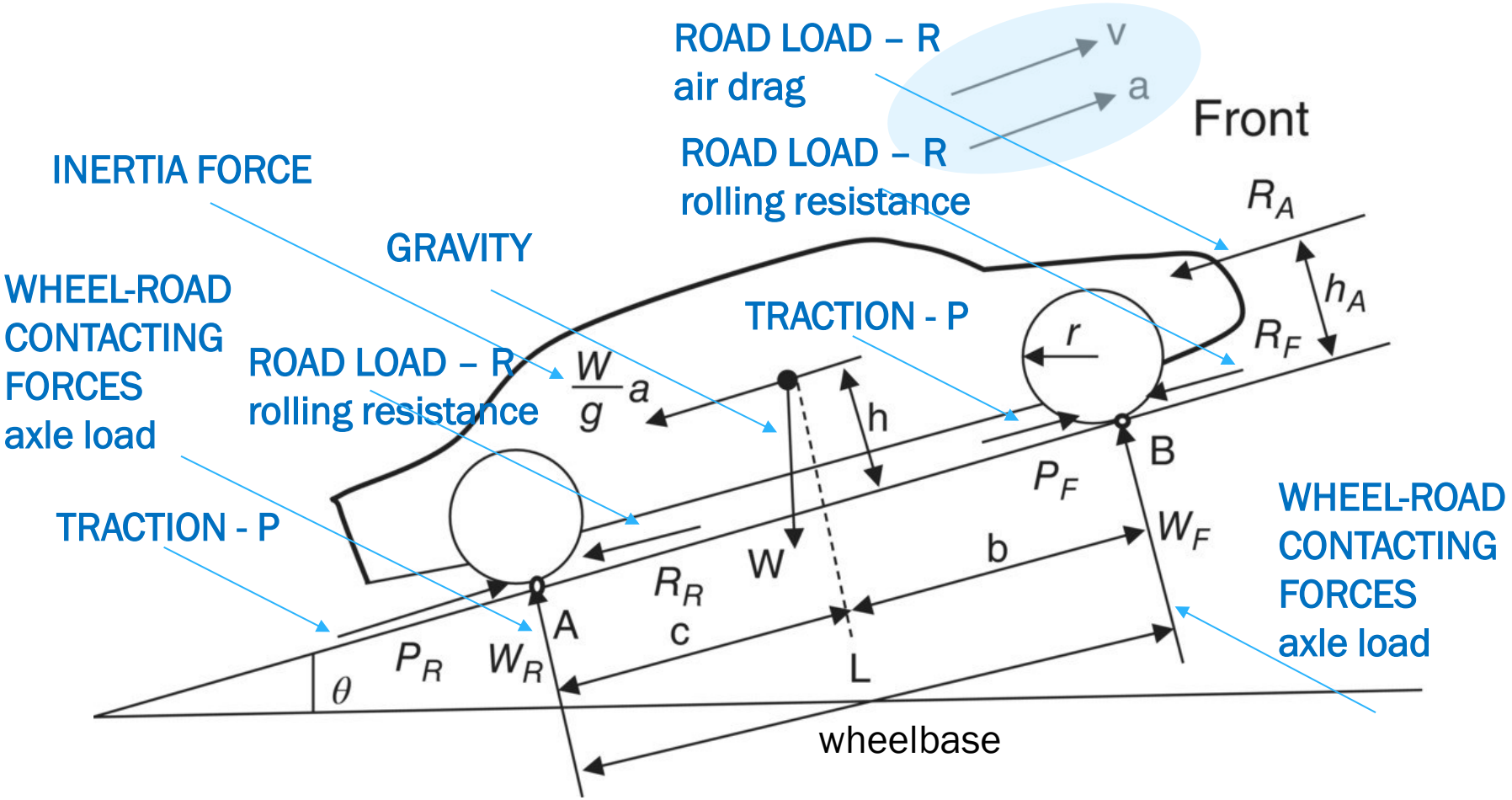
PYTHON PROJECT

FREE BODY DIAGRAM



PYTHON PROJECT

FREE BODY DIAGRAM



PYTHON PROJECT

OUR UNKNOWNS

- **WR(N)**
- **WF(N)**
- **WRstat(N)**
- **WFstat(N)**
- **a(m/s^2)**
- **P(N)**
- **R(N)**
- **HP(hp)**
- **Te(Nm)**
- **omega_e(RPM)**

PYTHON PROJECT

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PYTHON PROJECT

ROAD LOAD

$R_A = \frac{1}{2}\rho C_D A(v)^2$

air drag

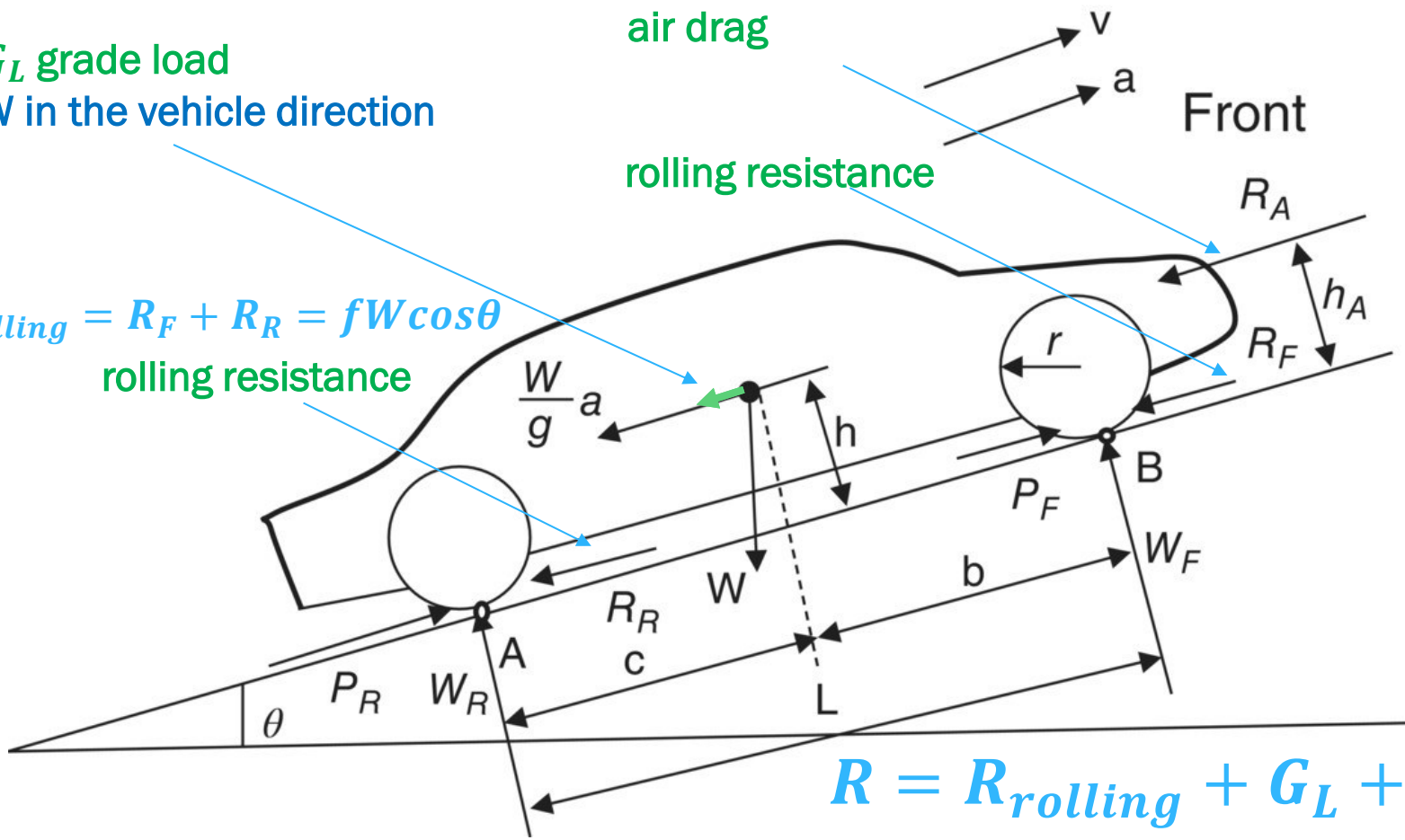
G_L grade load

W in the vehicle direction

rolling resistance

$R_{rolling} = R_F + R_R = fW\cos\theta$

rolling resistance



$R = R_{rolling} + G_L + R_A$

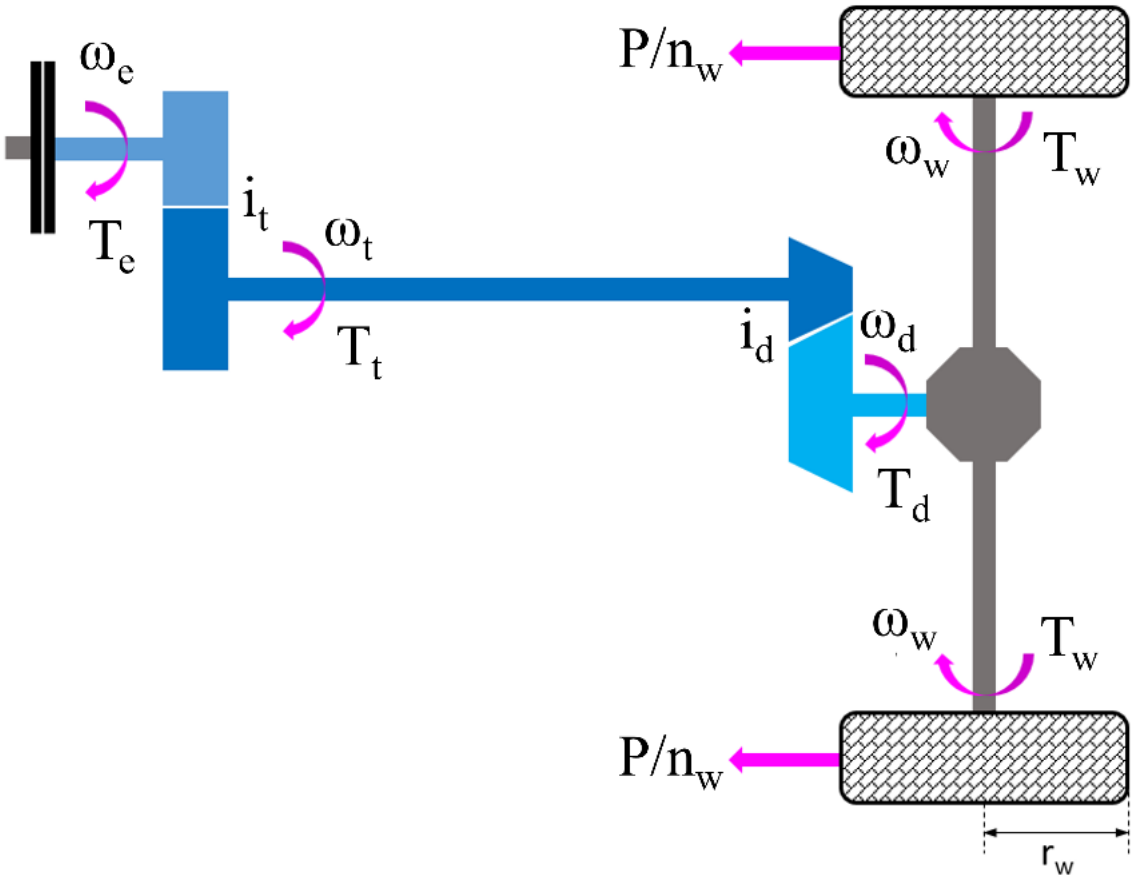
PYTHON PROJECT

OUR UNKNOWNS

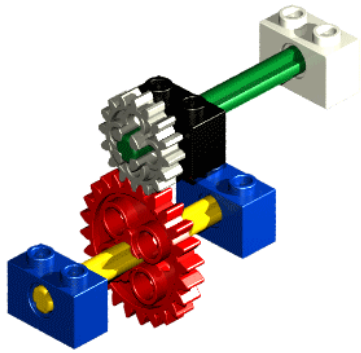
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PYTHON PROJECT

FINAL DRIVE

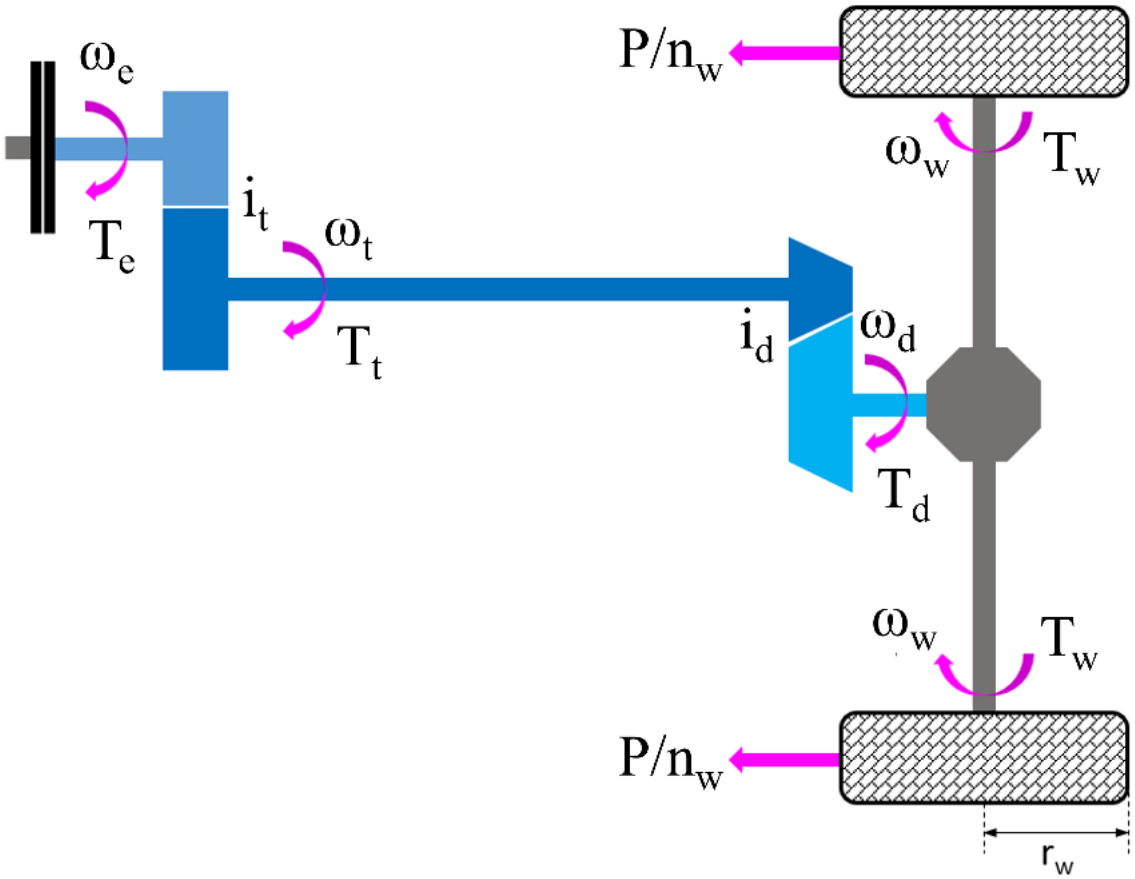


$$\omega_W = \frac{v}{r}$$



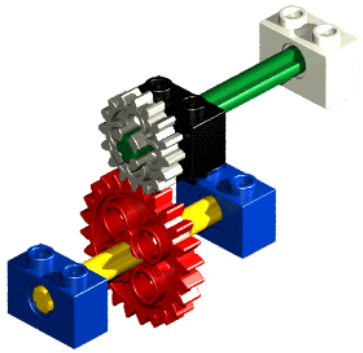
PYTHON PROJECT

FINAL DRIVE



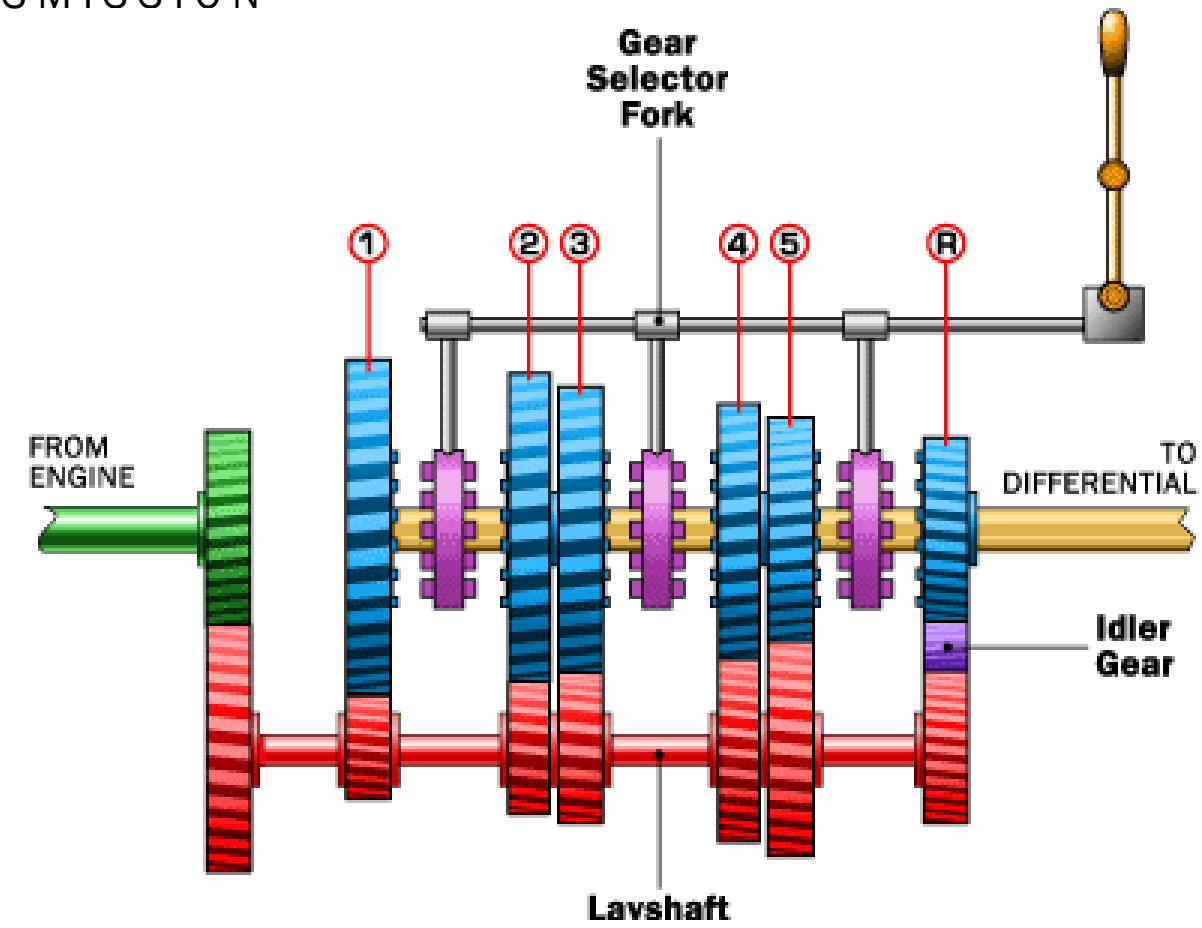
$$i_d = \frac{\omega_t}{\omega_W}$$

final drive ratio



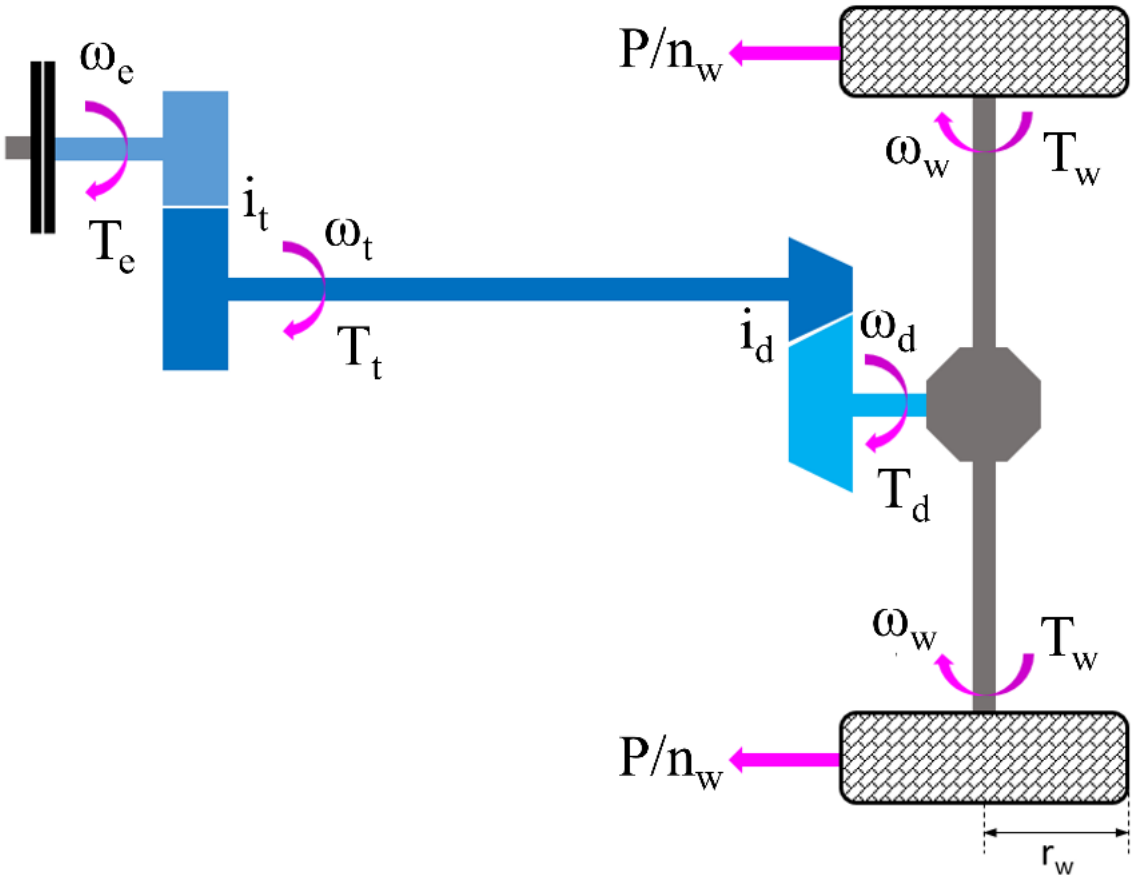
PYTHON PROJECT

TRASMISSION



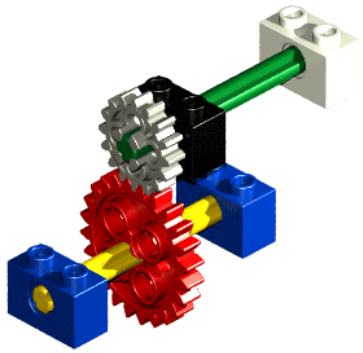
PYTHON PROJECT

TRANSMISSION

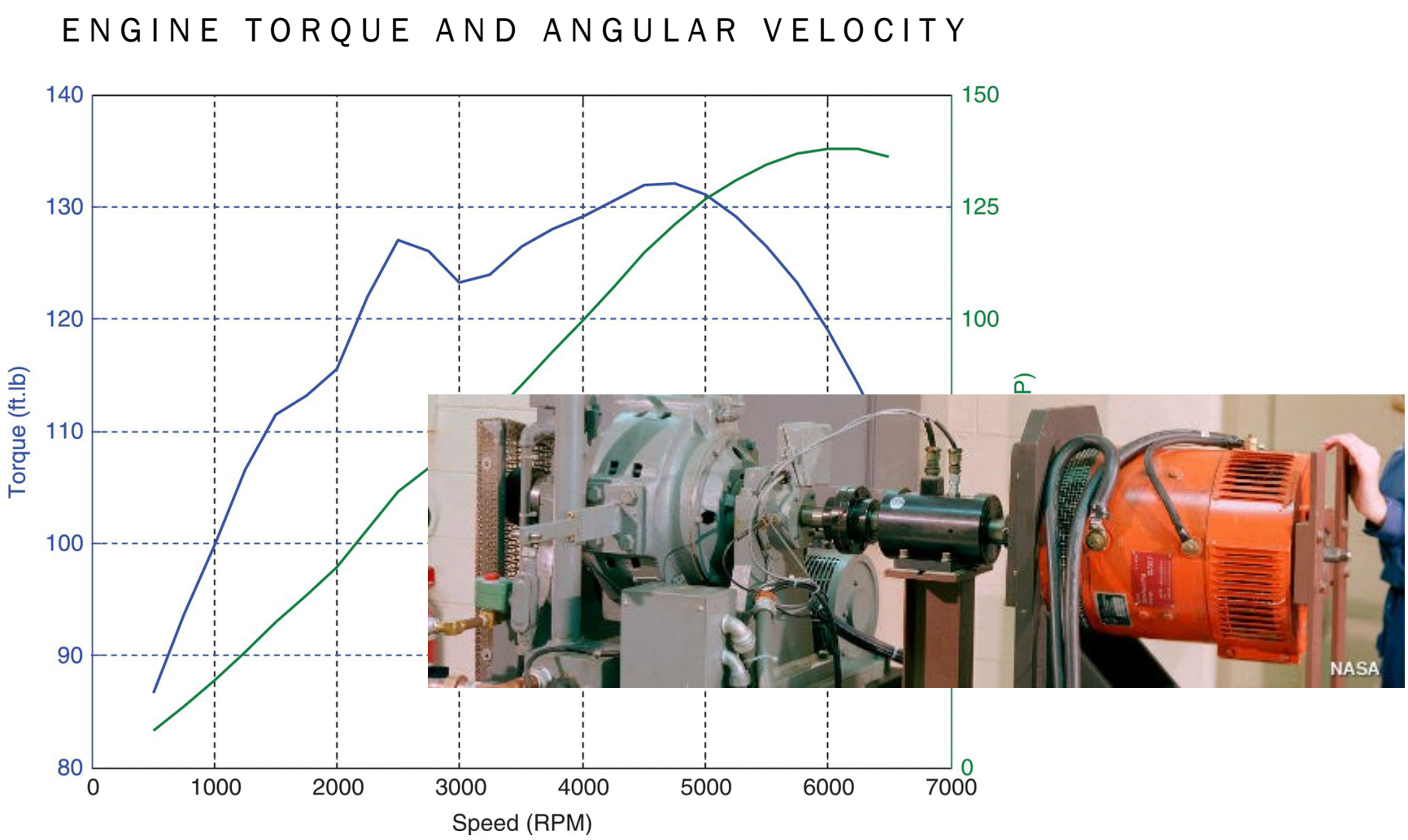


$$i_t = \frac{\omega_e}{\omega_t}$$

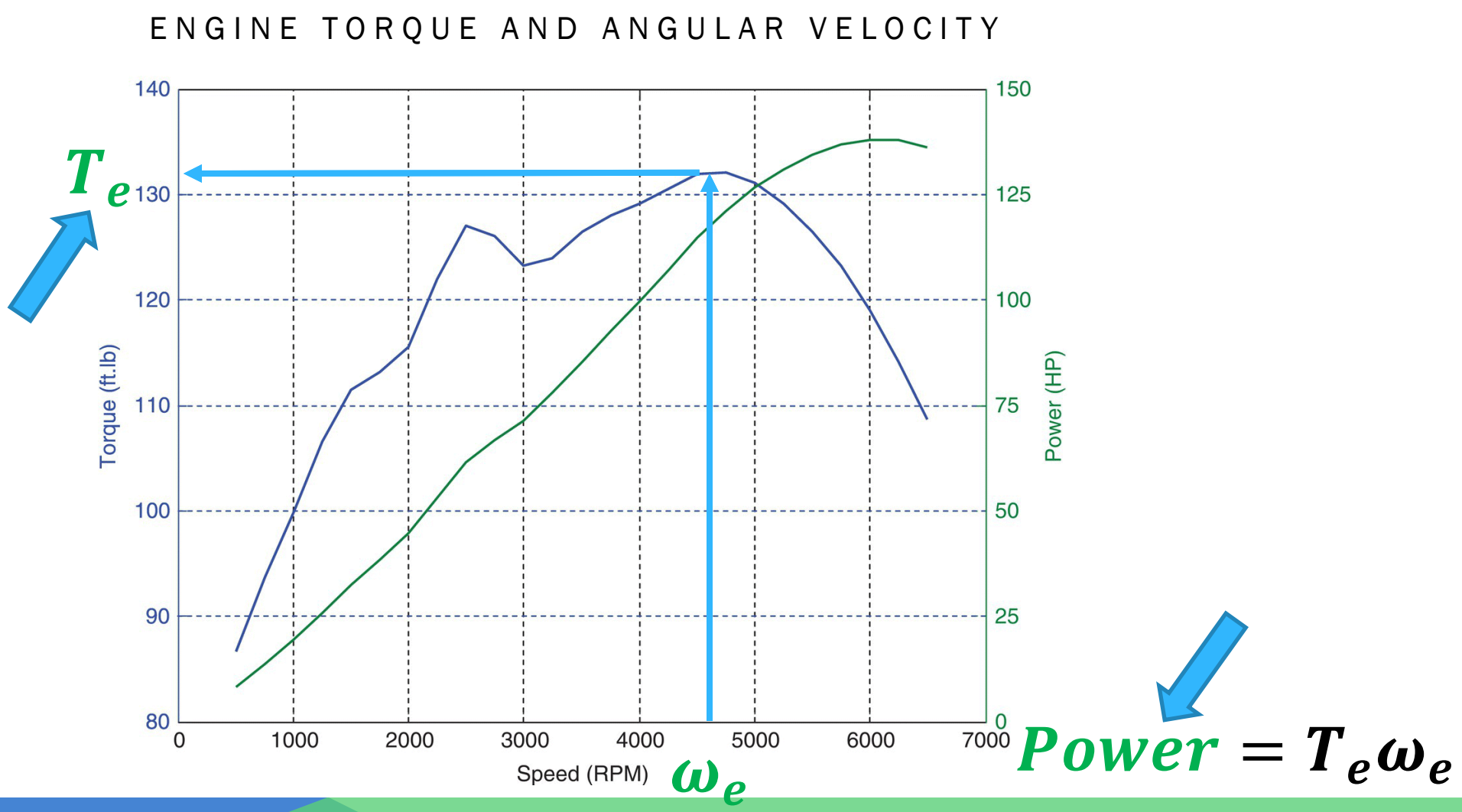
gear ratio



PYTHON PROJECT

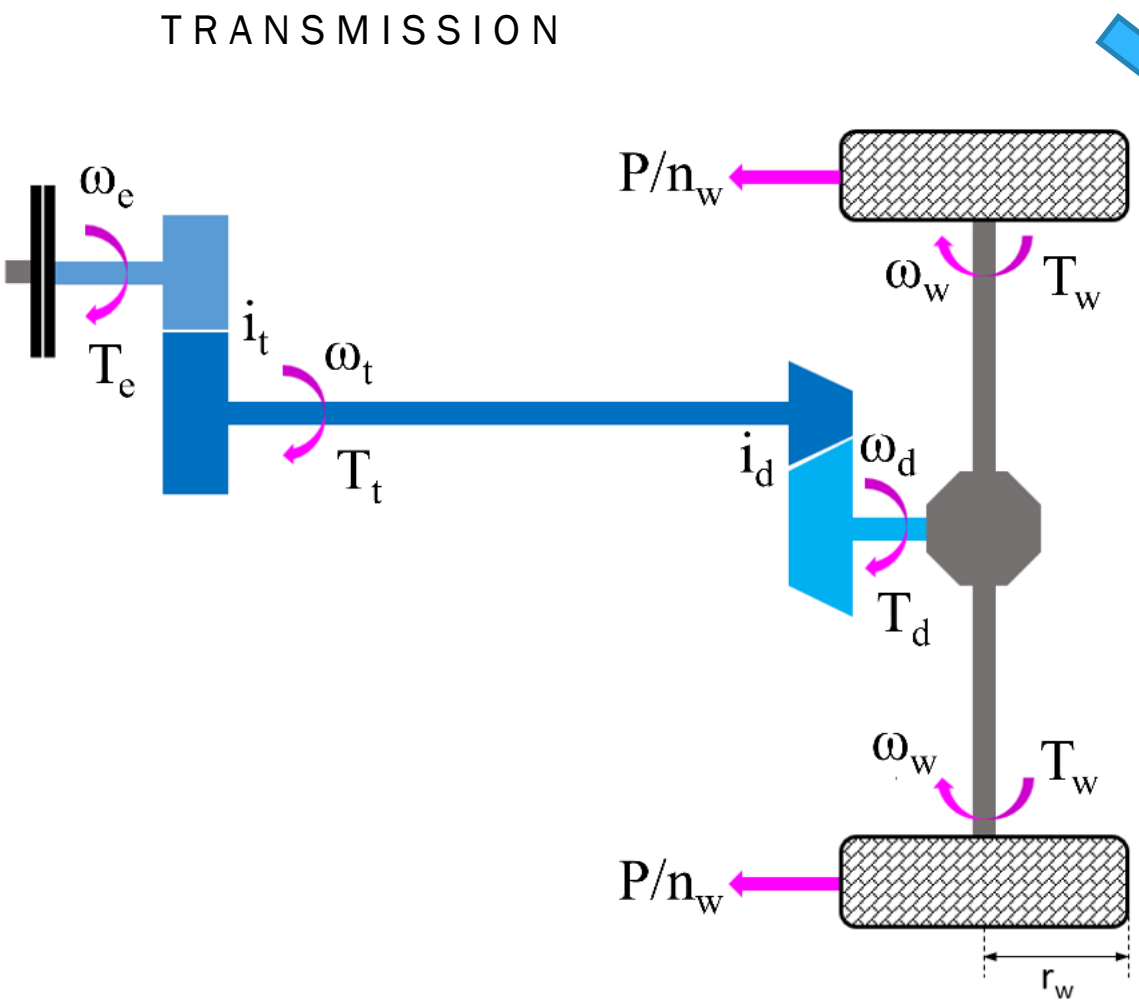


PYTHON PROJECT



PYTHON PROJECT

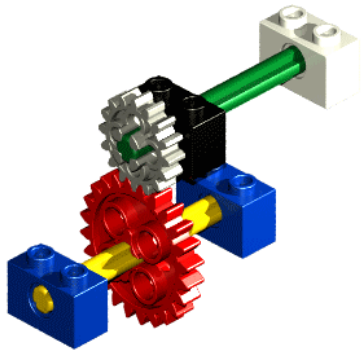
TRANSMISSION



transmission efficiency

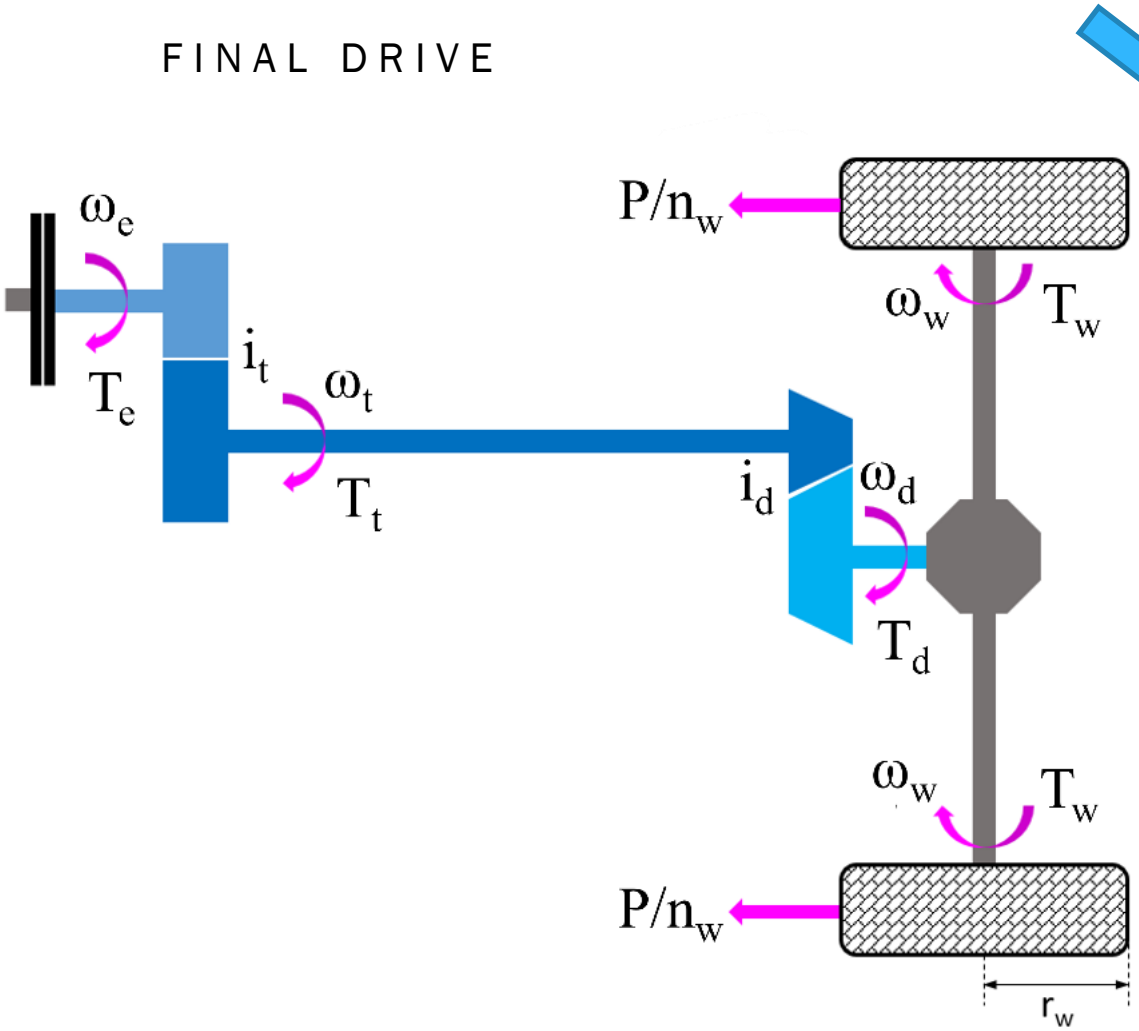
$$T_t = \eta_t i_t T_e$$

gear ratio



PYTHON PROJECT

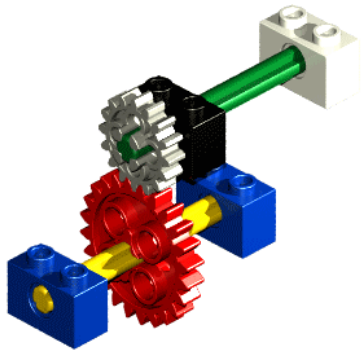
FINAL DRIVE



final drive efficiency

$$T_d = \eta_d i_d T_t$$

final drive ratio



PYTHON PROJECT

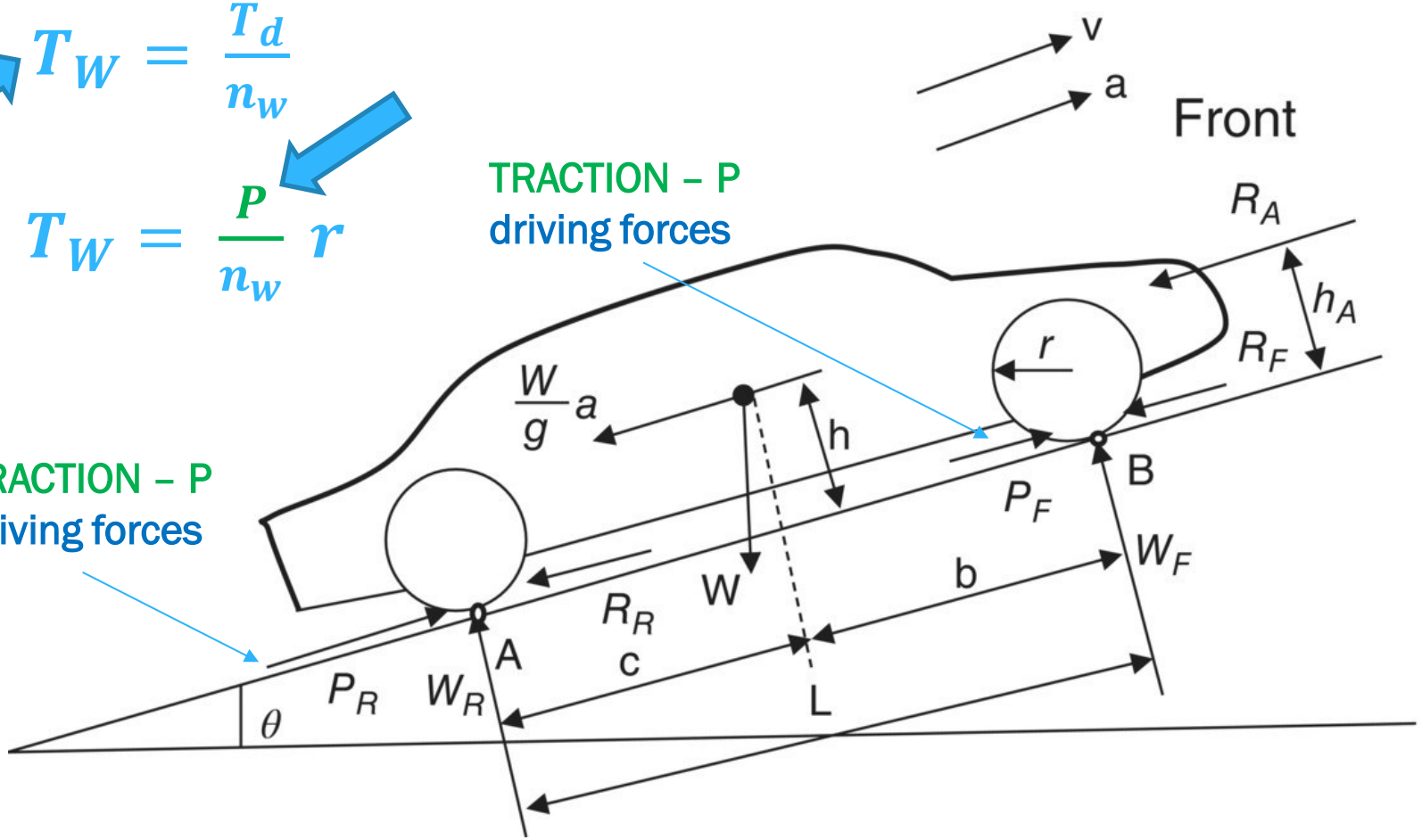
TRACTION

$$T_W = \frac{T_d}{n_w}$$

$$T_W = \frac{P}{n_w} r$$

TRACTION – P
driving forces

TRACTION – P
driving forces



PYTHON PROJECT

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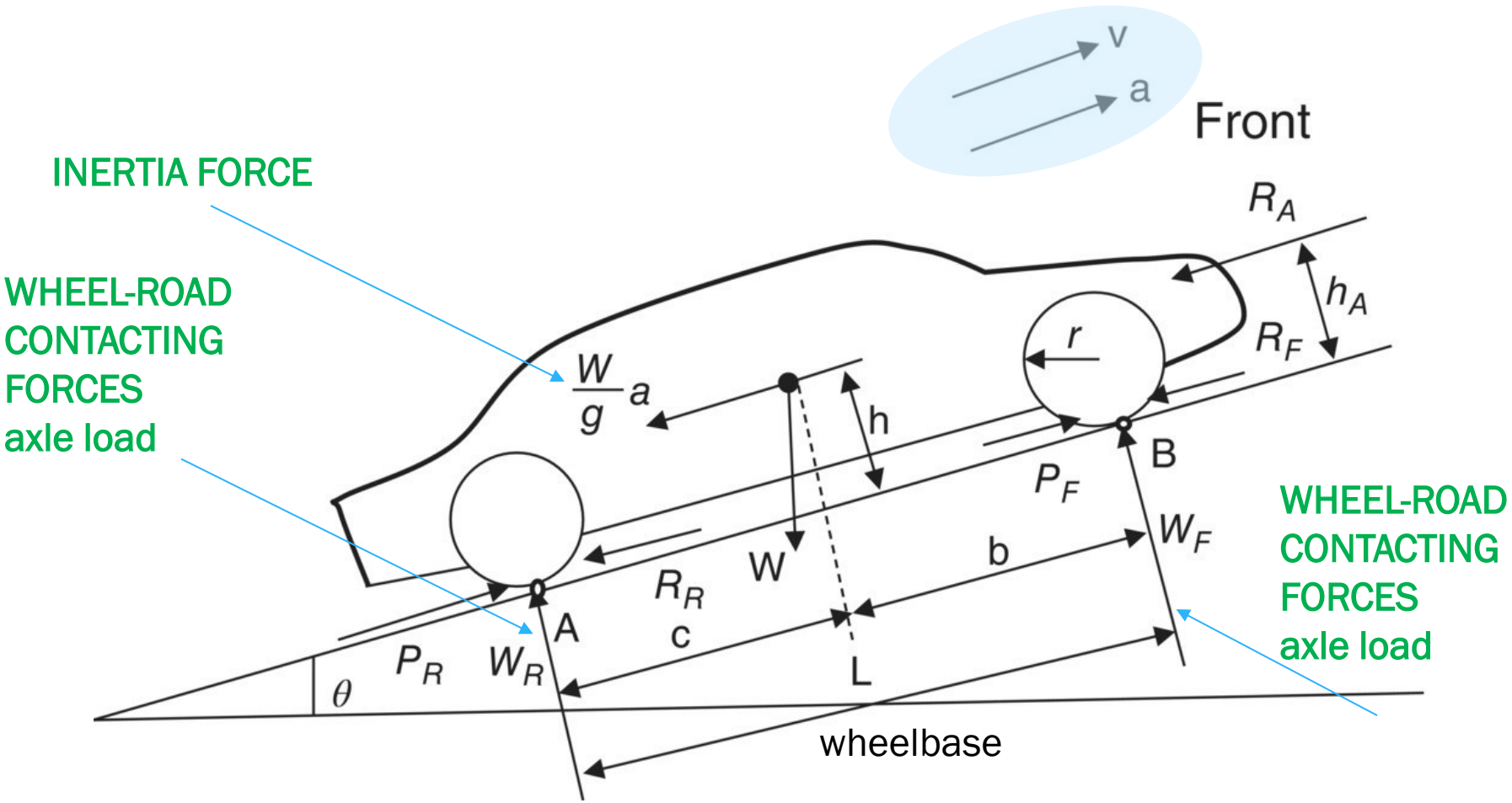
PYTHON PROJECT

OUR UNKNOWNS

- WR(N)
- WF(N)
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- WFstat(N)
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PYTHON PROJECT

AXLE LOADS AND ACCELERATION



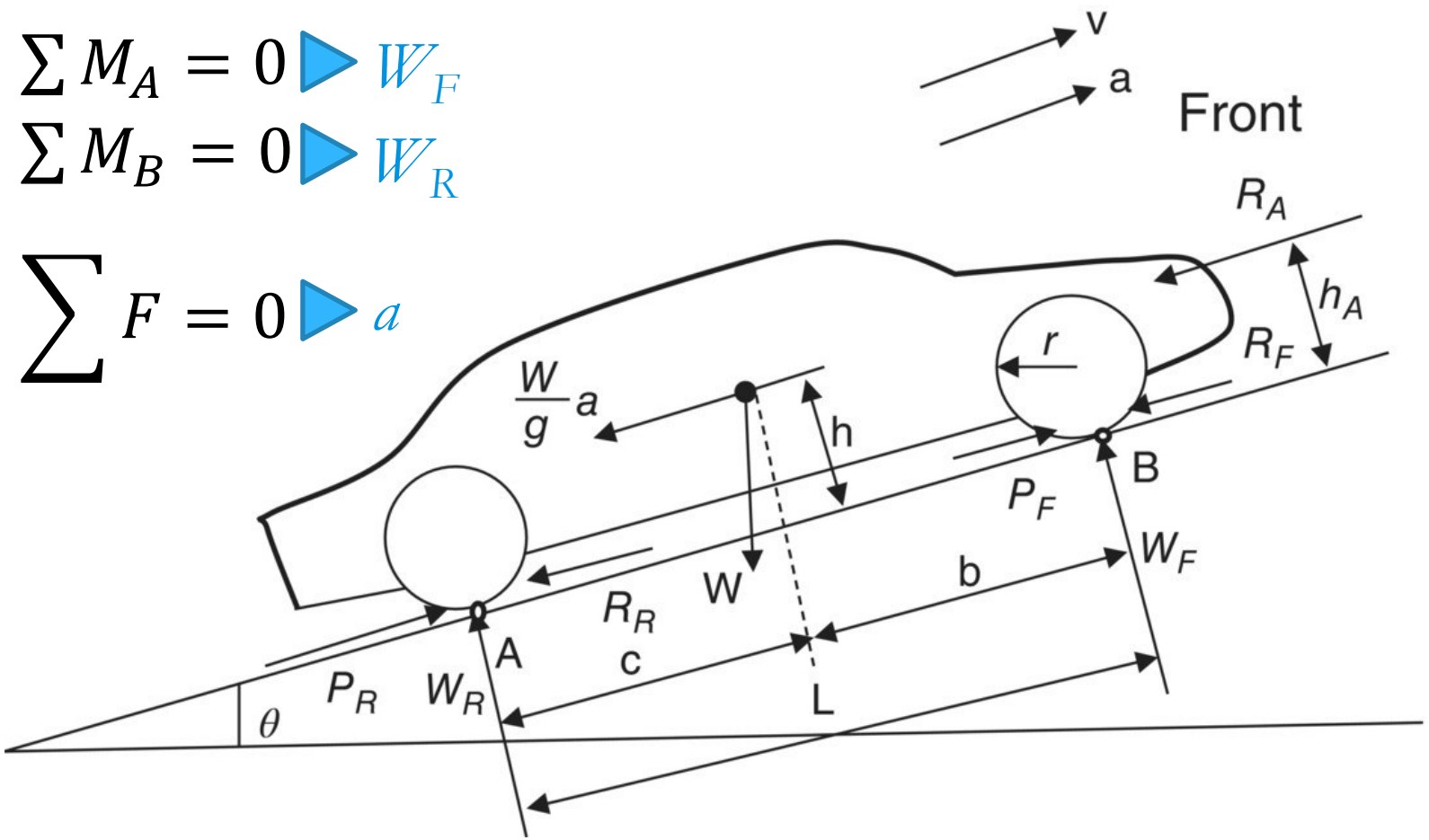
PYTHON PROJECT

FREE BODY DIAGRAM EQUILIBRIUM

$$\sum M_A = 0 \blacktriangleright W_F$$

$$\sum M_B = 0 \blacktriangleright W_R$$

$$\sum F = 0 \blacktriangleright a$$



PYTHON PROJECT

OUR UNKNOWNS

- `WR(N)`
- `WF(N)`
- `WRstat(N)`
- `WFstat(N)`
- `a(m/s²)`
- `P(N)`
- `R(N)`
- `HP(hp)`
- `Te(Nm)`
- `omega_e(RPM)`

static axle weights on level ground are obtained from the moment balance equations by making the slope angle θ, the air drag R_A, and the acceleration a equal to zero

PYTHON PROJECT

OUR UNKNOWNS

- WR(N)
- WF(N)
- WRstat(N)
- WFstat(N)
- a(m/s^2)
- P(N)
- R(N)
- HP(hp)
- Te(Nm)
- omega_e(RPM)

PYTHON PROJECT

OUR DATA: THE EXAMPLE CARS



PYTHON PROJECT

OUR DATA: THE EXAMPLE CARS



PYTHON PROJECT

OUR DATA: THE EXAMPLE CARS



PYTHON PROJECT

OUR DATA: THE EXAMPLE CARS



PYTHON PROJECT

OUR DATA

2001 AP1 Honda S2000

CAR DATA		
GEAR RATIOS	**value**	**test**
1st gear	3.133	3.133
2nd gear	2.045	2.045
3rd gear	1.481	1.481
4th gear	1.161	1.161
5th gear	0.971	0.971
6th gear	0.811	0.811



PYTHON PROJECT

OUR DATA

OTHER PARAMETERS	symbol	value	test
speed in km/h	v	100.000	100.000
terrain slope as %	slope %	2.000	2.000
wheelbase in m	L	2.400	2.400
radius of the wheel in m	r	0.406	0.406
roll resistance coefficient	f	0.018	0.018
air resistance moment arm, center of gravity height in m	hA,h	0.495	0.495
final drive efficiency as %	ηd	95.000	95.000
transmission efficiency as %	ηt	96.000	96.000
weight of the vehicle in N	W	12446.600	12446.600
air density in kg/m^3	ρ	1.230	1.230
drive ratio (final drive)	id	4.100	4.100
distance identifying the center of gravity position from the front wheel in m	b	1.540	1.540
drag coefficient	CD	0.330	0.330
frontal area in m^2	A	1.870	1.870



PYTHON PROJECT

OUR DATA

DYNAMOMETER DATA

ωe (RPM)	Te (N m)
1800.00	176.00
2200.00	185.20
2600.00	192.40
3000.00	199.30
3400.00	206.70
3800.00	214.90
4200.00	219.60
4600.00	222.20
5000.00	224.00
5400.00	228.30
5800.00	232.00
6200.00	234.30
6600.00	235.20
7000.00	234.00
7400.00	230.10

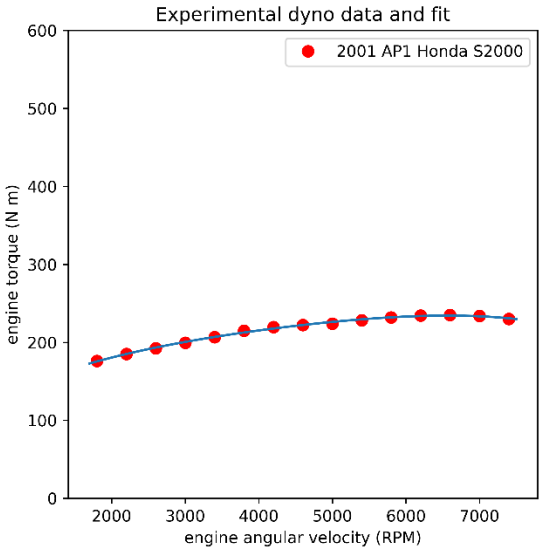
MAX ANGULAR VELOCITY

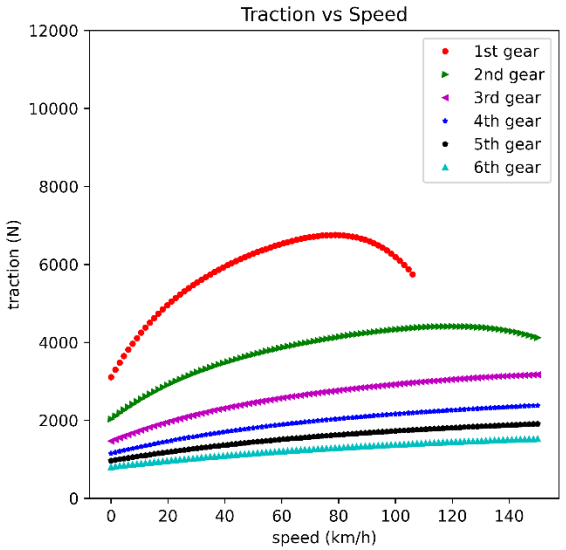
ωmax (RPM)	**9000**

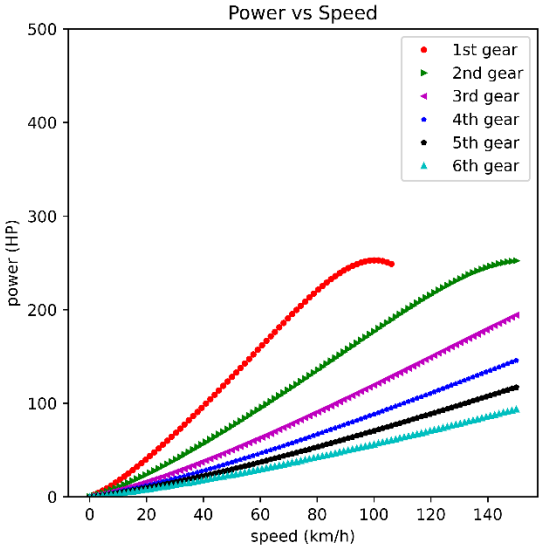
PYTHON PROJECT

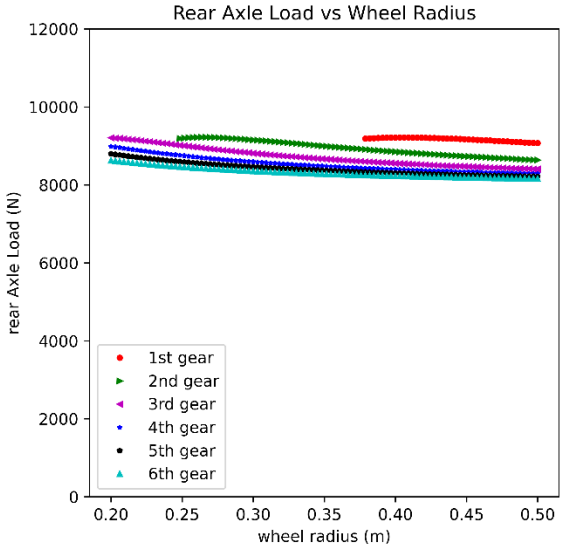
PLOT EXAMPLES







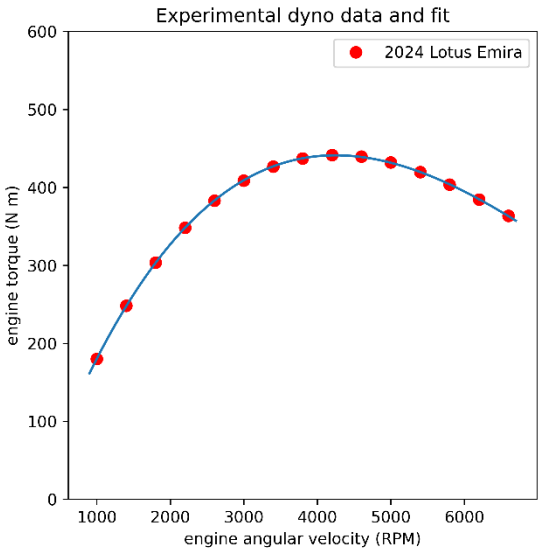


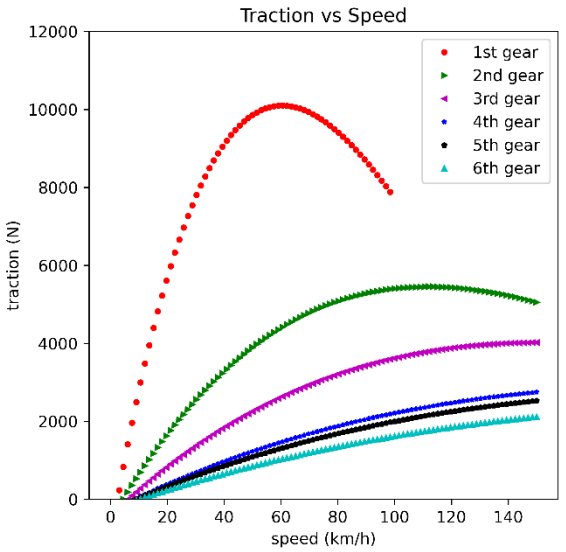


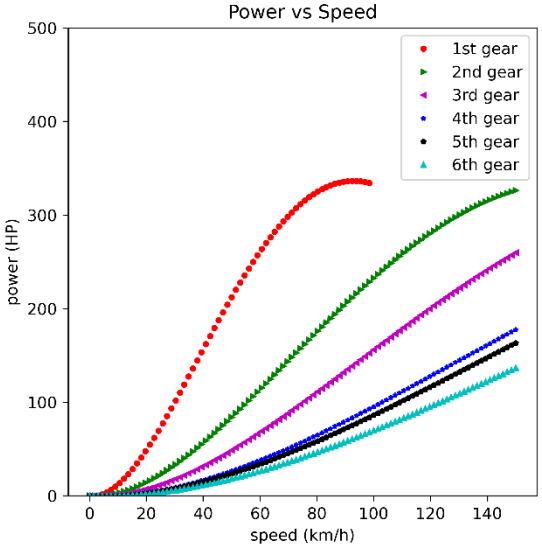
PYTHON PROJECT

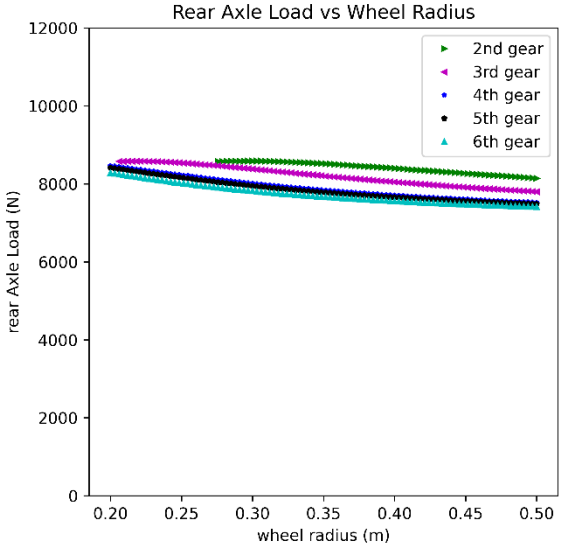
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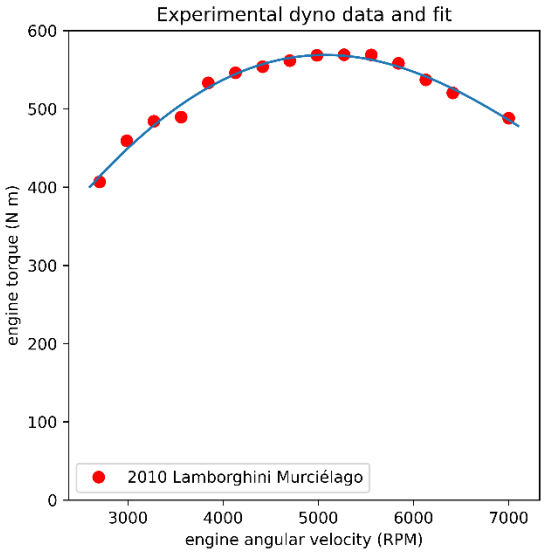


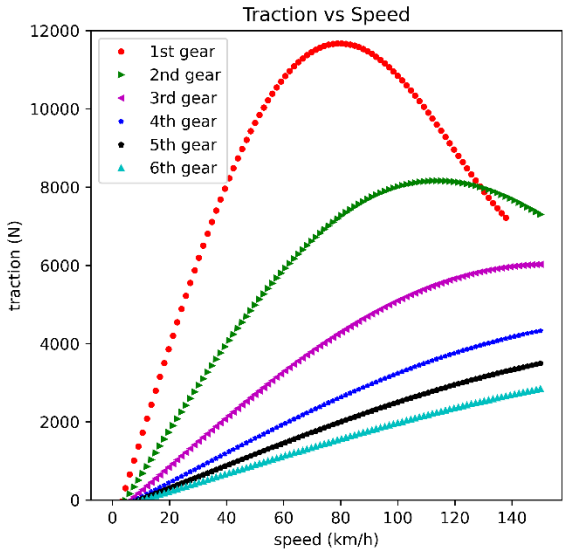


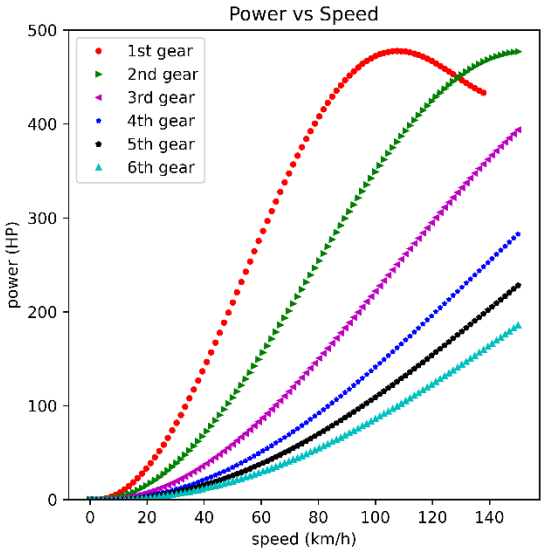
PYTHON PROJECT

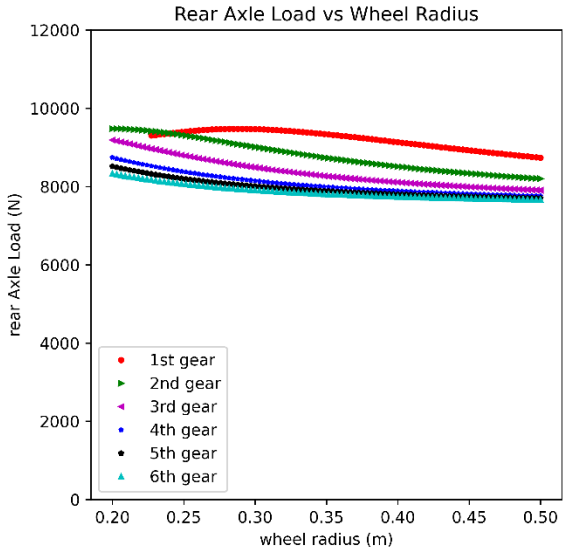
PLOT EXAMPLES







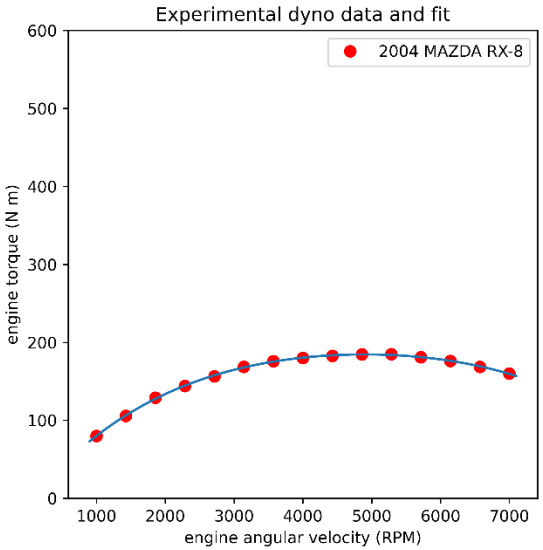


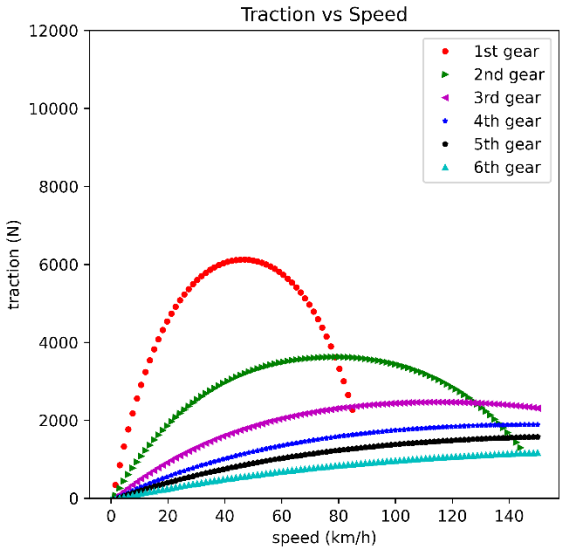


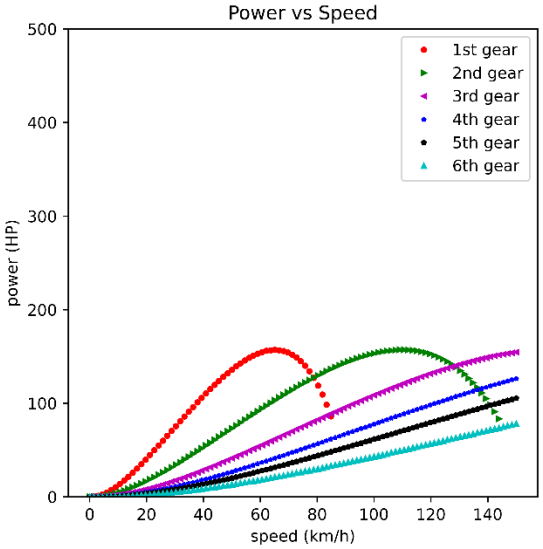
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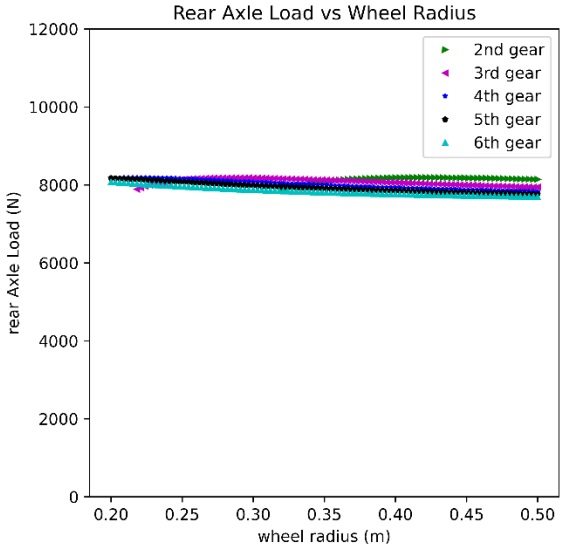
PLOT EXAMPLES





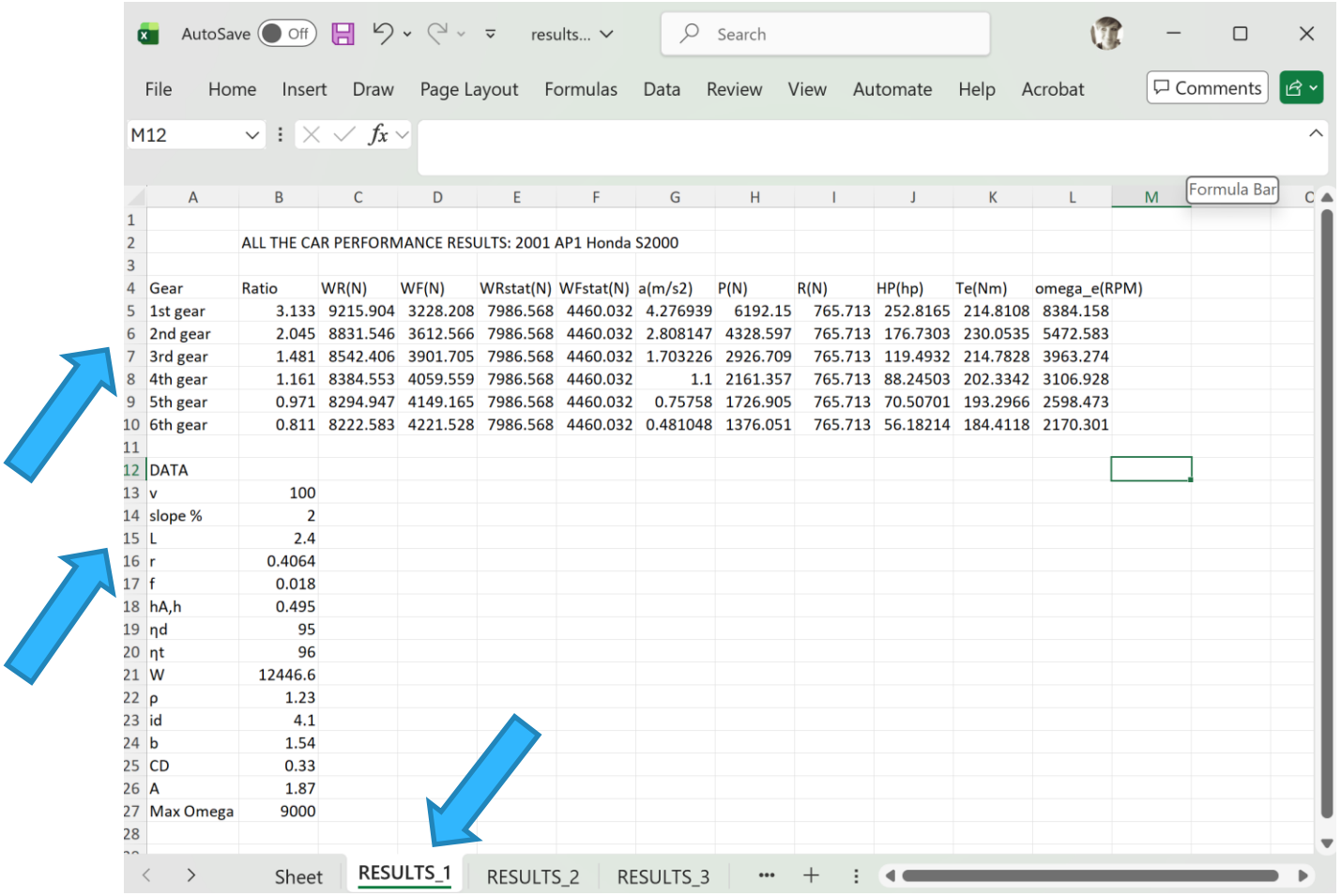






PYTHON PROJECT

FILE EXAMPLE



AutoSave Off | results... | Search | File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat | Comments | M12 | Formula Bar

ALL THE CAR PERFORMANCE RESULTS: 2001 AP1 Honda S2000

Gear	Ratio	WR(N)	WF(N)	WRstat(N)	WFstat(N)	a(m/s2)	P(N)	R(N)	HP(hp)	Te(Nm)	omega_e(RPM)
1st gear	3.133	9215.904	3228.208	7986.568	4460.032	4.276939	6192.15	765.713	252.8165	214.8108	8384.158
2nd gear	2.045	8831.546	3612.566	7986.568	4460.032	2.808147	4328.597	765.713	176.7303	230.0535	5472.583
3rd gear	1.481	8542.406	3901.705	7986.568	4460.032	1.703226	2926.709	765.713	119.4932	214.7828	3963.274
4th gear	1.161	8384.553	4059.559	7986.568	4460.032	1.1	2161.357	765.713	88.24503	202.3342	3106.928
5th gear	0.971	8294.947	4149.165	7986.568	4460.032	0.75758	1726.905	765.713	70.50701	193.2966	2598.473
6th gear	0.811	8222.583	4221.528	7986.568	4460.032	0.481048	1376.051	765.713	56.18214	184.4118	2170.301

DATA	
v	100
slope %	2
L	2.4
r	0.4064
f	0.018
hA,h	0.495
ηd	95
ηt	96
W	12446.6
ρ	1.23
id	4.1
b	1.54
CD	0.33
A	1.87
Max Omega	9000

Sheet | RESULTS_1 | RESULTS_2 | RESULTS_3

PYTHON PROJECT

```
MAIN MENU

**************************************************
                CAR PERFORMANCE ANALYSIS
**************************************************
Choose the vehicle type:
1. 2001 AP1 Honda S2000
2. 2024 Lotus Emira
3. 2010 Lamborghini Murciélago
4. 2004 MAZDA RX-8
**************************************************
5. EXIT
**************************************************
```

PYTHON PROJECT



```
Select a car type (1-4) or EXIT (5): 1

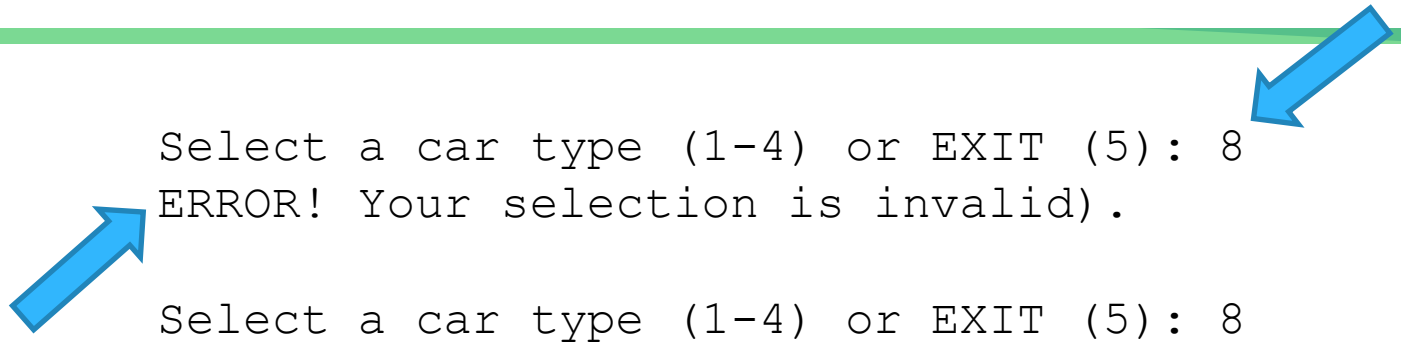
**************************************************
          CHECKING THE ORIGINAL DATA FILE
**************************************************
All data are positive.


Select a car type (1-4) or EXIT (5): 2

**************************************************
          CHECKING THE ORIGINAL DATA FILE
**************************************************
All data are positive.
```



PYTHON PROJECT



```
Select a car type (1-4) or EXIT (5): 8
ERROR! Your selection is invalid).

Select a car type (1-4) or EXIT (5): 8
ERROR! Your selection is invalid).

Select a car type (1-4) or EXIT (5): 2

**************************************************
          CHECKING THE ORIGINAL DATA FILE
**************************************************
All data are positive.
```

PYTHON PROJECT

```
**************************************************
                CAR PERFORMANCE ANALYSIS
**************************************************
Choose the vehicle type:
1. 2001 AP1 Honda S2000
2. 2024 Lotus Emira
3. 2010 Lamborghini Murciélago
4. 2004 MAZDA RX-8
**************************************************
5. EXIT
**************************************************


Select a car type (1-4) or EXIT (5): 5

**************************************************
                    THANK YOU.
**************************************************
```



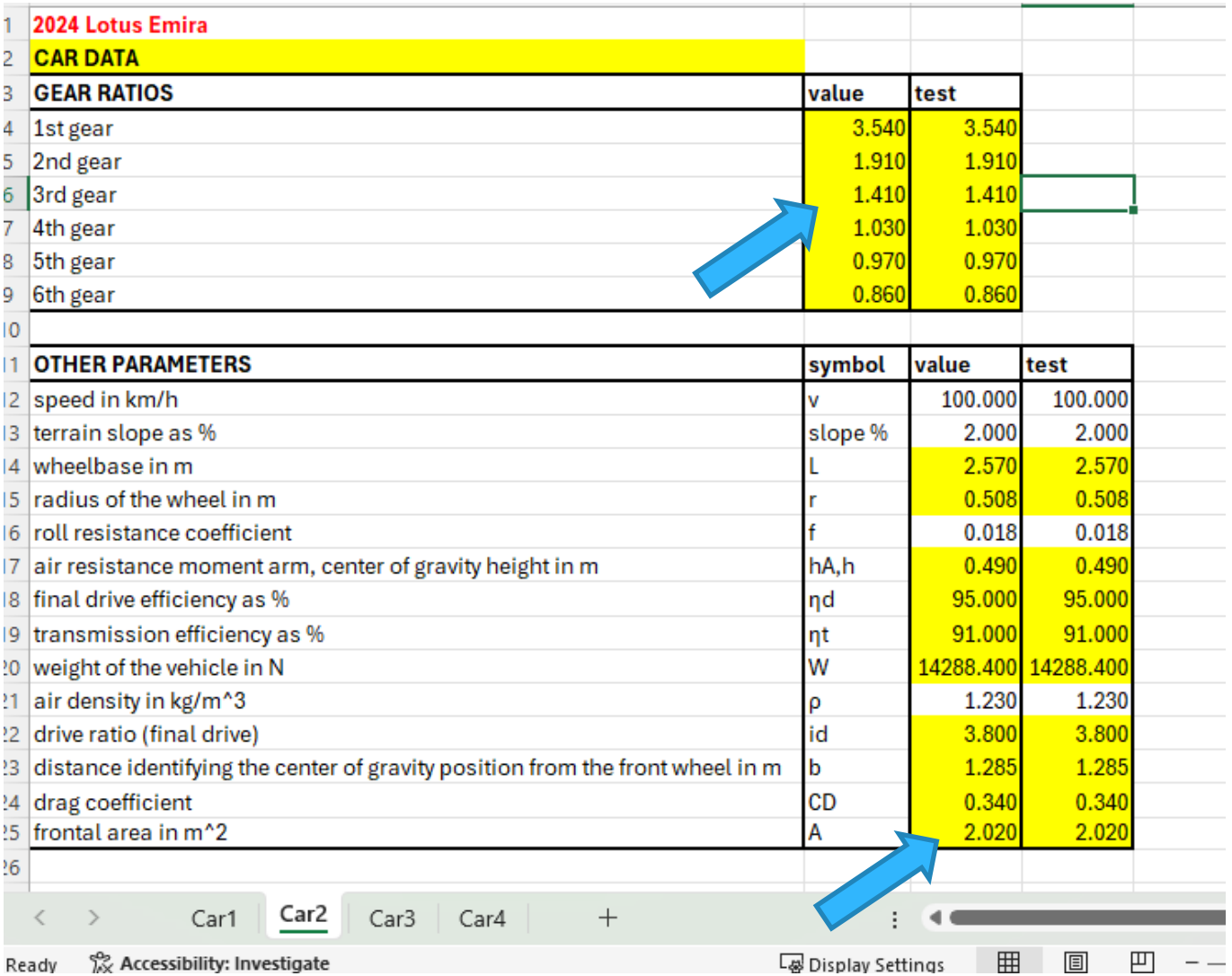
PYTHON PROJECT

```
Select a car type (1-4) or EXIT (5): 2

**************************************************
          CHECKING THE ORIGINAL DATA FILE
**************************************************
The 1st gear is negative. Please insert it again: -1
The 1st gear is negative. Please insert it again: 3.54
The slope % is negative. Please insert it again: -1
The slope % is negative. Please insert it again: -2
The slope % is negative. Please insert it again: 2
All data are positive.
```



PYTHON PROJECT



1	2024 Lotus Emira			
2	CAR DATA			
3	GEAR RATIOS	value	test	
4	1st gear	3.540	3.540	
5	2nd gear	1.910	1.910	
6	3rd gear	1.410	1.410	
7	4th gear	1.030	1.030	
8	5th gear	0.970	0.970	
9	6th gear	0.860	0.860	
10				
11	OTHER PARAMETERS	symbol	value	test
12	speed in km/h	v	100.000	100.000
13	terrain slope as %	slope %	2.000	2.000
14	wheelbase in m	L	2.570	2.570
15	radius of the wheel in m	r	0.508	0.508
16	roll resistance coefficient	f	0.018	0.018
17	air resistance moment arm, center of gravity height in m	hA,h	0.490	0.490
18	final drive efficiency as %	ηd	95.000	95.000
19	transmission efficiency as %	ηt	91.000	91.000
20	weight of the vehicle in N	W	14288.400	14288.400
21	air density in kg/m^3	ρ	1.230	1.230
22	drive ratio (final drive)	id	3.800	3.800
23	distance identifying the center of gravity position from the front wheel in m	b	1.285	1.285
24	drag coefficient	CD	0.340	0.340
25	frontal area in m^2	A	2.020	2.020
26				

PYTHON PROJECT

SUGGESTIONS:

Create first a function that calculates all the unknowns for <u>one</u> set of data (e.g. one speed) and <u>one</u> gear ratio

When everything is OK, then extend to all the gears:

```
for gear in gears:
        use the function you created
```

Plot one point at time, so you can perform the OMEGA MAX check:

```
for gear in gears:
      for each_point in all_points:
            use the function you created
```

PYTHON PROJECT

SUGGESTIONS:

```
cell_address = sheet["A1"]
actual_cell_value = cell_address.value
```



```
actual_cell_value = (sheet["A{}".format(i)]).value
```

For the new tabs:

```
# Create a new worksheet
sheet_new = wb_results.create_sheet(sheet_name)
```